SHASHI BHUSHAN JHA

Communication Systems Researcher | Quantum Communication & Future Networks

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RESEARCH PROFILE

M.Tech graduate in Electrical Engineering (Communication and Signal Processing) with completed research in analytical modeling and simulation of wireless systems. My thesis combined mathematical derivations with Monte Carlo validation for multi-user NOMA, including BER, throughput, and SIC error modeling. My primary doctoral interest is quantum communication and quantum-safe networking; I am building the required foundation in quantum information, channels, QKD, and quantum networks while retaining complementary interests across future communication systems.

RESEARCH INTERESTS

- **Primary:** quantum communication and quantum information, including state representation and quantum-channel modeling
- Quantum key distribution, quantum networks and repeaters, error correction, and hybrid PQC–QKD security
- 6G and beyond systems, optical communication, ISAC, NTN, and satellite communication
- Multiple access, NOMA, PHY-layer signal processing, and analytical performance evaluation

EDUCATION

Master of Technology in Electrical Engineering — 2024–2026
Indian Institute of Technology Ropar — *CGPA: 7.78/10*
Specialization: Communication and Signal Processing
Relevant Coursework: RF Systems for Communication; Analytical Techniques for Signal Processing; Detection and Estimation Theory
Thesis: "Generalization of Downlink and Uplink NOMA Framework for Future Generation Technology"
Supervisor: Dr. Satyam Agarwal

Bachelor of Engineering in Electrical Engineering — 2016–2022
Purbanchal University, Biratnagar, Nepal — *CGPA: 2.98/4.00*
Major: Power Systems

MANUSCRIPT IN PREPARATION

Analytical BER Study with Hardware Validation for N-User NOMA — **Internal review**
Prepared for submission to an IEEE Transactions journal; currently undergoing internal faculty review.

RESEARCH EXPERIENCE

Master's Thesis Research — Power-Domain NOMA — Jul 2025–May 2026
Indian Institute of Technology Ropar — *Communication and Signal Processing*

- Developed a unified BER and throughput framework for 2–4-user uplink and downlink power-domain NOMA over flat Rayleigh fading with explicit SIC-error modeling.
- Derived closed-form two-user downlink BER expressions and SIC-consistent uplink and generalized multi-user formulations.
- Evaluated power allocation, distance-based user ordering, SIC depth, NOMA–OMA throughput, and oversampling through Monte Carlo simulation.
- Showed how oversampling reduces effective noise variance while user ordering and error propagation continue to govern SIC reliability.

Research Intern — Autonomous Systems — Sep 2025–Mar 2026
SkyFlock Uaviation Pvt. Ltd. — *Remote*

- Studied swarm-drone architectures and inter-drone communication protocols.
- Used ROS 2, Gazebo, PX4, and MAVSDK for multi-agent simulation, control, and telemetry analysis.

PROFESSIONAL EXPERIENCE

Trainee Associate — Prosecution and Drafting Department 18 May 2026–12 Aug 2026

InventIP Legal Services LLP *Gurugram*

- Worked in the Prosecution and Drafting Department, gaining professional experience with patent prosecution and drafting workflows.

Instructor — Computer Science and Mathematics Apr 2022–Dec 2023

Delhi Public School, Dharan *Nepal*

- Taught Python programming, computational thinking, and mathematics; developed course materials and mentored senior students.

SELECTED TECHNICAL PROJECTS

Rat-Race Hybrid Coupler Course project

- Designed and simulated a rat-race hybrid coupler in Ansys HFSS for the RF Systems for Communication course.

Echo Cancellation Using Adaptive Filtering Jan–May 2025

- Implemented normalized least-mean-square adaptive filtering in MATLAB to suppress acoustic echo and assess convergence and signal-quality behavior.

Audio Signal Analysis 2024

FFT, spectrogram, and MFCC-based time-frequency analysis using Python, SciPy, and Librosa.

Smart Surveillance 2024

Real-time motion detection and object tracking using Python, OpenCV, edge detection, and frame differencing.

TECHNICAL SKILLS

Wireless and DSP	Power-domain NOMA, SIC, BER analysis, Rayleigh fading, communication theory, detection and estimation, adaptive filtering
Research Methods	Analytical modeling, Monte Carlo simulation, performance characterization, technical writing, literature review
Programming	MATLAB, Python, C++; NumPy, Pandas, SciPy, scikit-learn, OpenCV
Tools and Platforms	Ansys HFSS, Linux, Git/GitHub, ROS 2, Gazebo, PX4, MAVSDK, Docker

ADDITIONAL INFORMATION

Certifications	WIPO General Course on Intellectual Property; WIPO Essentials of Patents; CS105: Introduction to Python; Introduction to the Internet of Things
Languages	English (professional), Hindi (full professional), Nepali (native/bilingual), Maithili (native/bilingual)

Updated September 2026